Fuzzing for Input Validation Bugs

Step 2: Initial Execution of Buggy Application

The unpatched parser script was executed interactively using well-formed inputs (00:30, a05, and electric,45,yes). Under normal inputs, all three functions execute without raising exceptions.

Command: python bike_share_rental_parser.py

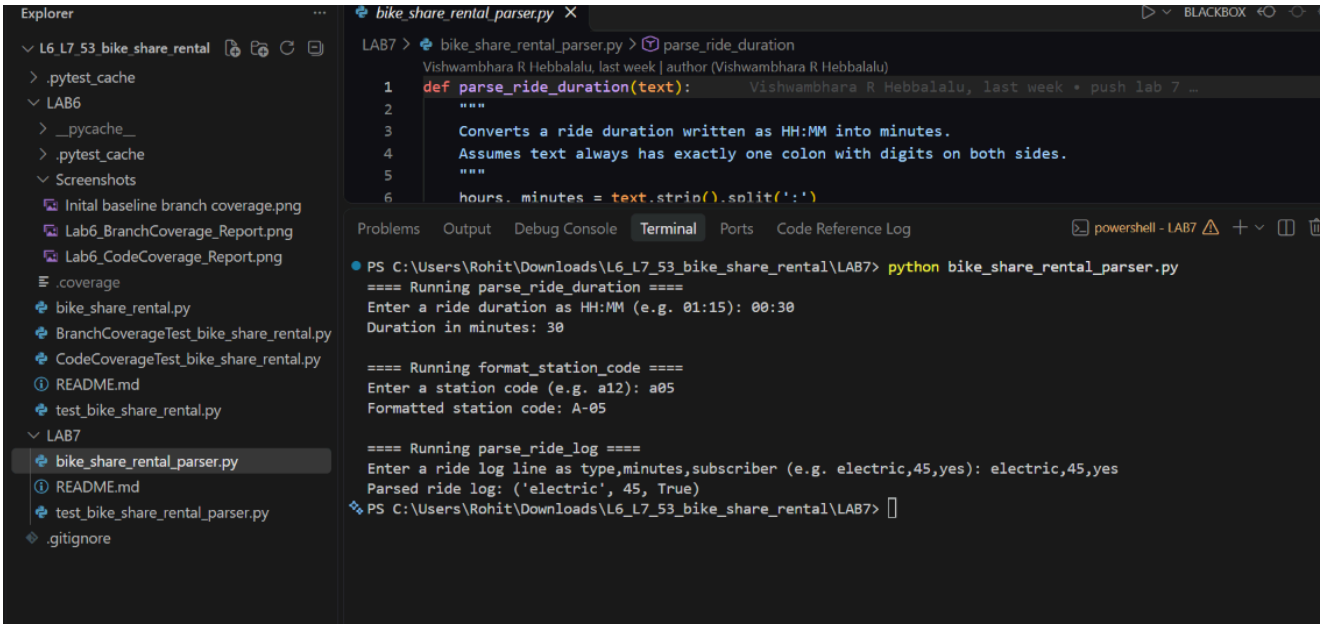


Figure 1: Baseline execution of bike_share_rental_parser.py with standard inputs.

Steps 3 & 4: Property-Based Fuzz Testing with Hypothesis (Failing Run)

Property-based tests using @given(st.text() | st.none()) were added to test_bike_share_rental_parser.py to verify that invalid inputs raise controlled ValueError exceptions rather than unhandled crashes. Running Pytest exposed unhandled AttributeError and IndexError crashes on edge cases such as None and empty strings.

Command: python -m pytest test_bike_share_rental_parser.py

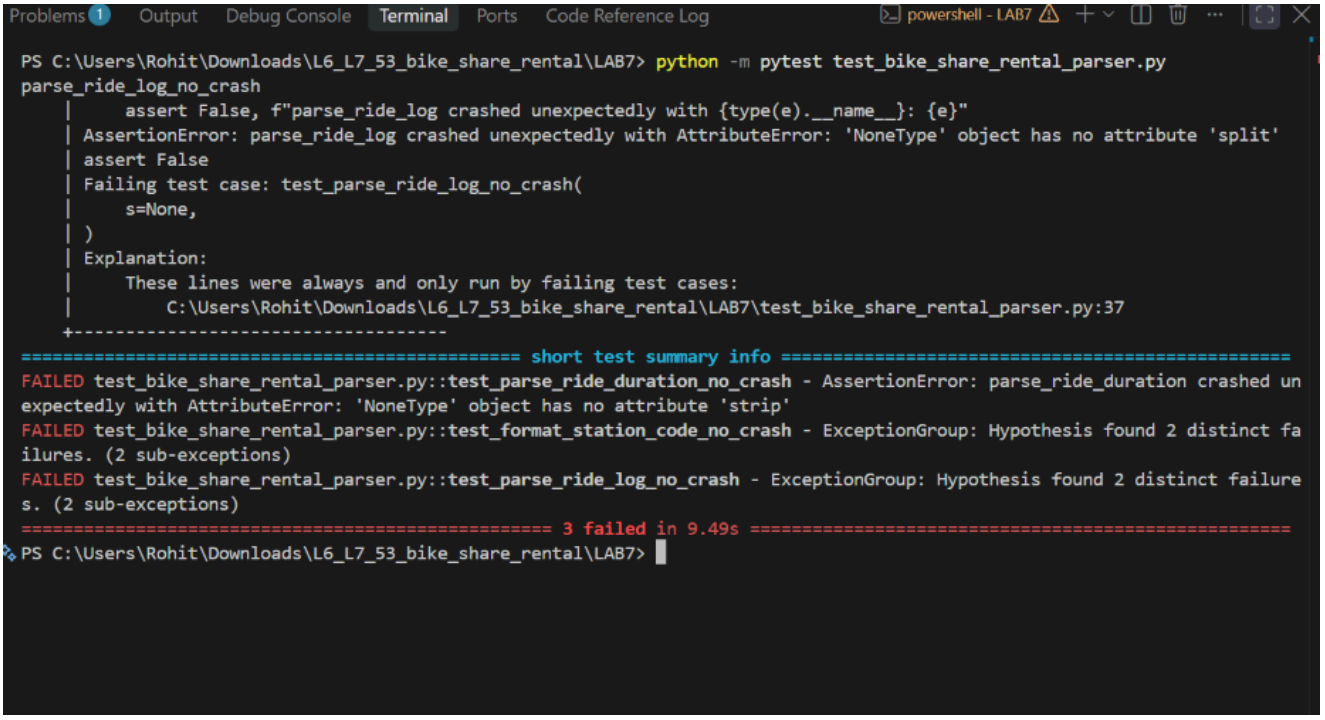


Figure 2: Hypothesis fuzzing run exposing 3 failing test cases and falsifying edge-case inputs.

Step 5: Execution of Hardened Application

Defensive input validation was implemented across parse_ride_duration, format_station_code, and parse_ride_log to enforce string types, delimiter counts, ASCII digit checks, and numeric bounds, raising descriptive ValueError messages on malformed data.

Command: python bike_share_rental_parser.py

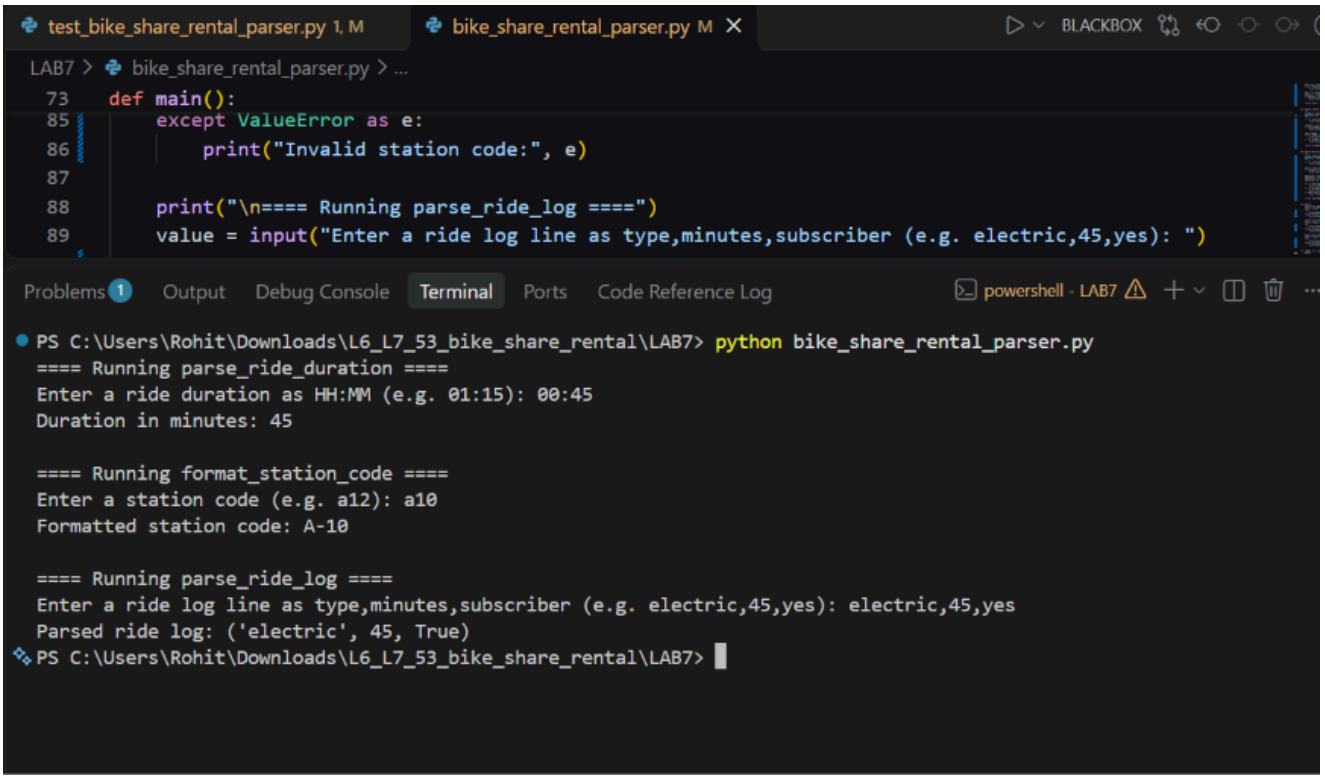


Figure 3: Execution of the hardened bike_share_rental_parser.py script.

Step 6: Re-Running Hypothesis Fuzz Tests on Fixed Script (Passing Run)

Re-running the Hypothesis test suite against the hardened parser confirms that all generated Unicode strings, empty inputs, and None values are handled safely without unexpected exceptions.

Command: python -m pytest test_bike_share_rental_parser.py

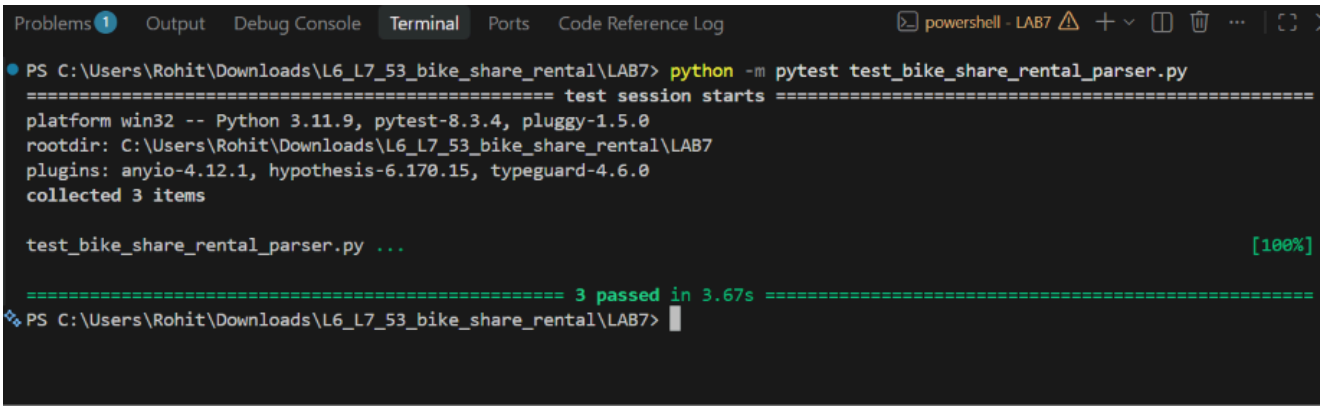


Figure 4: Verification run confirming all 3 Hypothesis property tests pass (3 passed in 3.67s).

Summary of Input Validation Fixes

parse_ride_duration(text)	AttributeError on None; ValueError on missing/extra colons, non-digits, or out-of-range minutes.	Verified str type, exact HH:MM split, ASCII digits, and 0 ≤ minutes < 60.
format_station_code(code)	AttributeError on None; IndexError on empty/whitespace strings; malformed single-char codes.	Verified str type, minimum length of 2, ASCII letter prefix, and ASCII digit suffix.
parse_ride_log(line)	AttributeError on None; IndexError when fewer than 3 fields; invalid minute values or subscriber flags.	Verified str type, exact 3 comma-separated fields, non empty bike type, digit minutes, and yes/no flag.